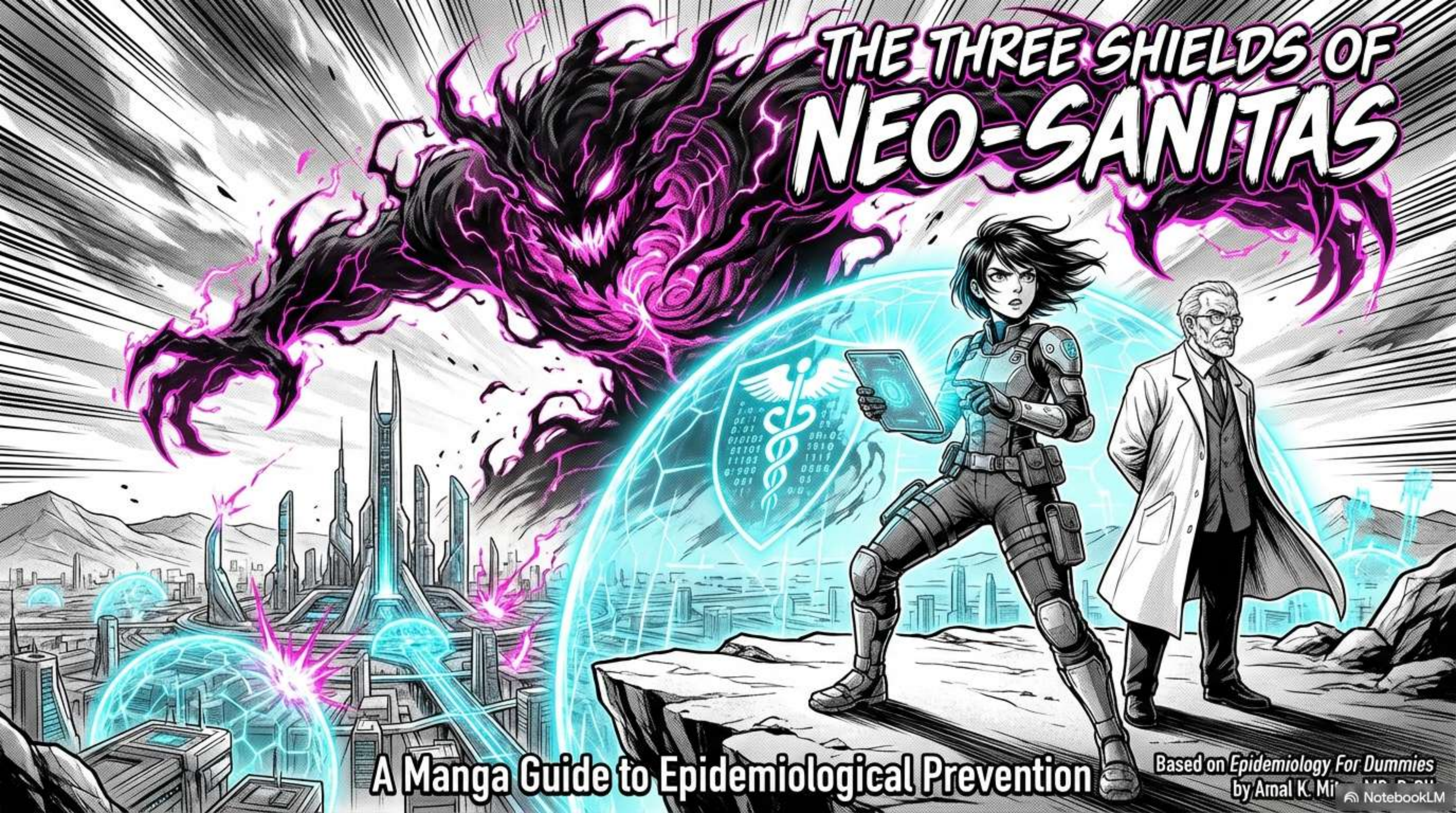


THE THREE SHIELDS OF NEO-SANITAS



A Manga Guide to Epidemiological Prevention

Based on *Epidemiology For Dummies*
by Amal K. Mi
NotebookLM



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PHYSICIAN VS. EPIDEMIOLOGIST

Physician:

- Focuses on treating/curing disease in the individual patient.
- Reactive (passive process).

Epidemiologist:

- Focuses on preventing illness in the community/population.
- Proactive (active process).

THE GOAL

Promote health and prevent illness before it strikes.



SHIELD 1: PRIMARY PREVENTION



Initiating
Phase One! Deploy
the Outer Barrier!

破境
THE BARRIER

Excellent. This is
Primary Prevention.
We stop the occurrence
of disease before
it begins.

SHIELD 1: PRIMARY PREVENTION

Definition: Measures taken to avert the occurrence of a disease in a person who is apparently healthy and does not yet have the disease.

Target: Healthy individuals.

Key Strategy: Reducing risk factors to prevent onset.





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THE BRICKS: VACCINES

- The single most important tool for primary prevention.
- **Vaccine-Preventable Diseases:** Measles, Polio, Tetanus, Diphtheria, Covid-19, Hepatitis B.

IMMUNITY TYPES

- **Natural Active:** From exposure to disease.
- **Artificial Active:** From vaccines (long-lasting).

NATURAL ACTIVE → 

ARTIFICIAL ACTIVE → 

HERD IMMUNITY

When a large proportion of a group is immune, it protects the minority who are not.

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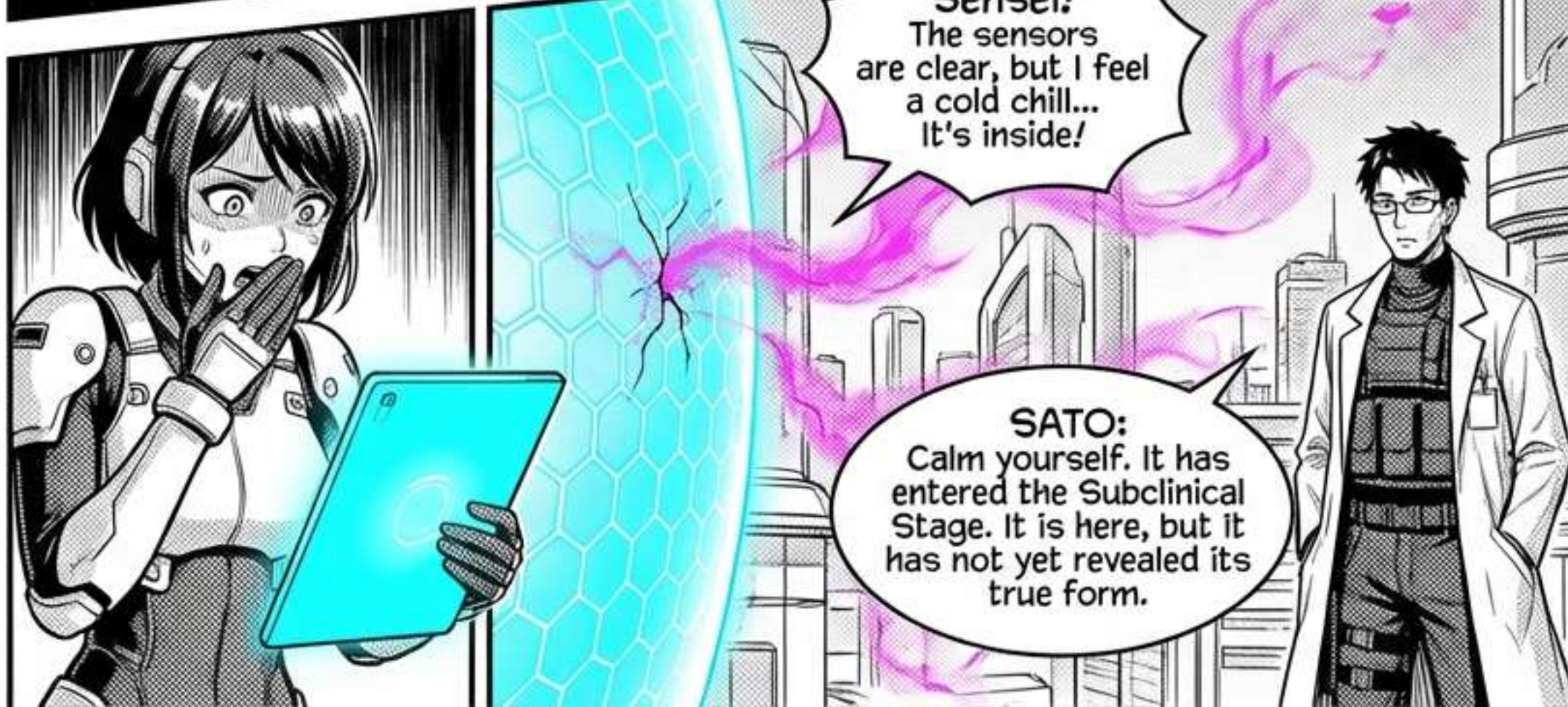


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FORTIFYING THE POPULATION

Health Promotion Strategies:

- **Health Education:** Hand washing, social distancing, mask-wearing (crucial for communicable diseases).
- **WASH:** Safe Water, Sanitation, and Hygiene.
- **Nutrition:** Balanced diet for immunity.
- **Prenatal Care:** Protecting the next generation.



Sensei!
The sensors
are clear, but I feel
a cold chill...
It's inside!

SATO:
Calm yourself. It has
entered the Subclinical
Stage. It is here, but it
has not yet revealed its
true form.

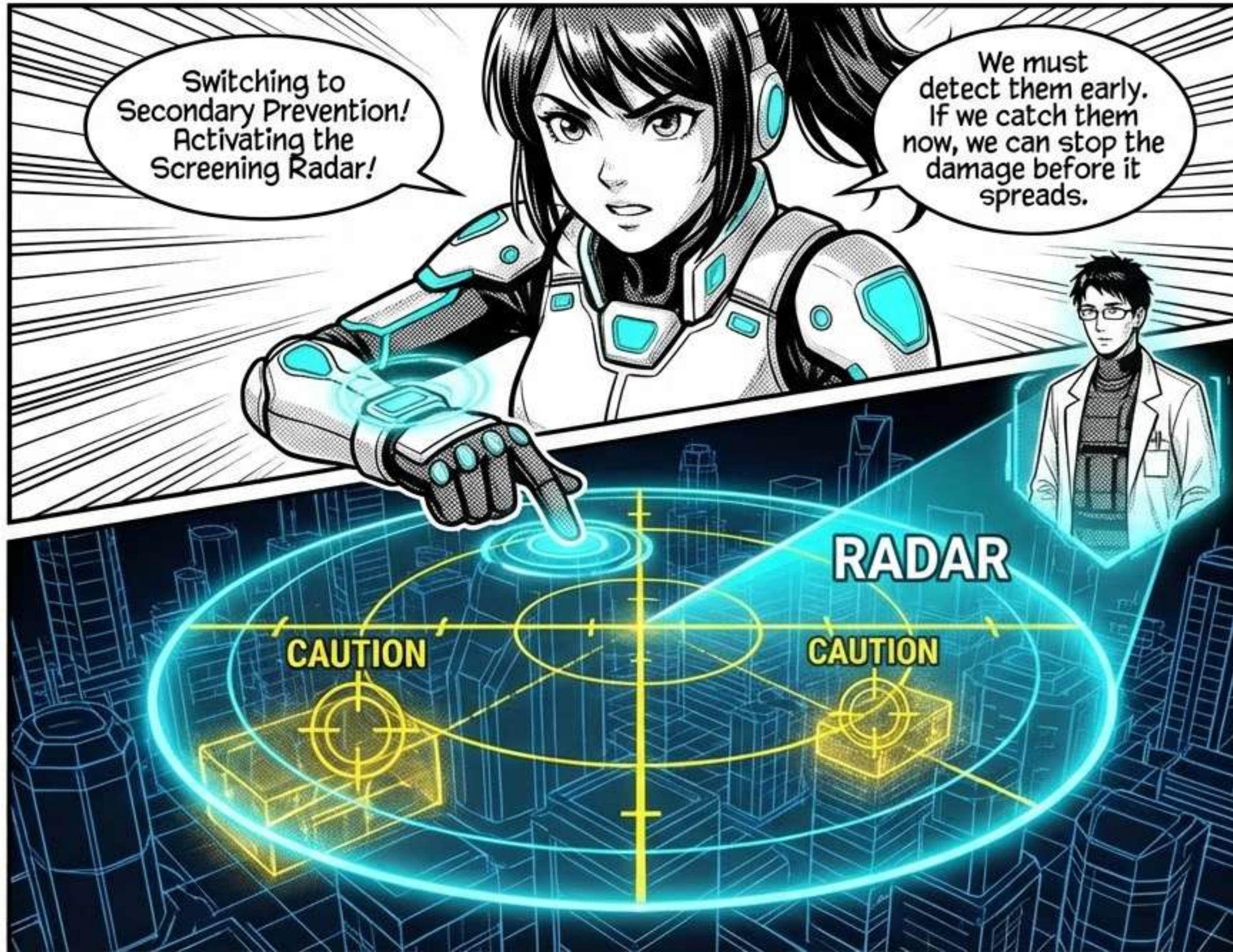
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THE BREACH: SUBCLINICAL STAGE

Definition: The time between infection/exposure and the appearance of clinical symptoms.

- Incubation Period (Infectious Diseases): e.g., Cholera (few hours to 3 days).
- Induction/Latency Period (Chronic Diseases): e.g., Cancer/HIV (months to years).

SHIELD 2: SECONDARY PREVENTION



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SHIELD 2: SECONDARY PREVENTION

Definition: Early detection of disease to reduce severity, duration, and complications.

Goal: Identify disease in the earliest stages (pre-clinical) before the onset of signs and symptoms.

Primary Tool: Screening.



Targets acquired!
The scanner is
picking up anomalies
in asymptomatic
citizens.

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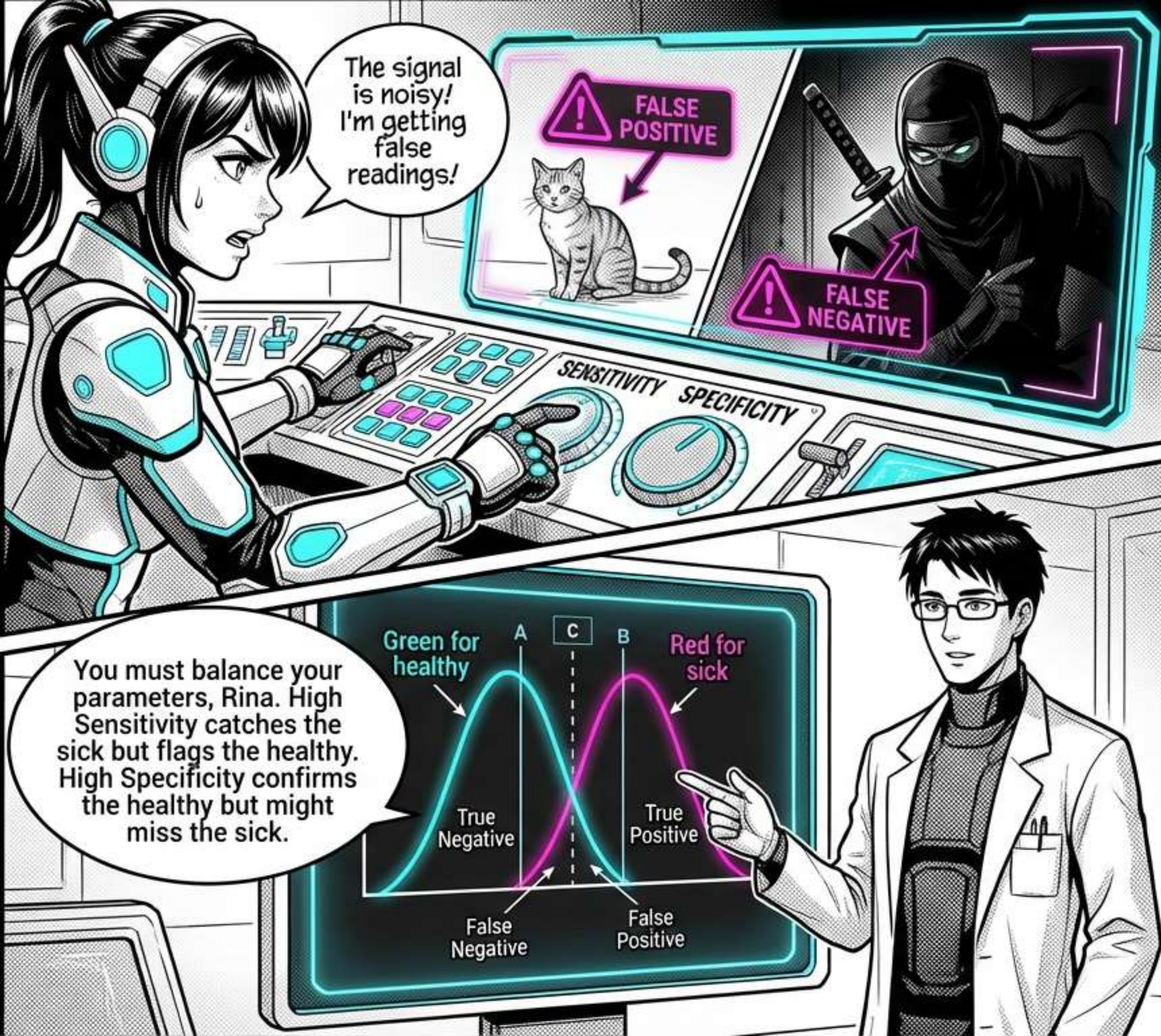
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SCREENING TECHNIQUES

Common Screening Examples:

- Mammograms/Self-exams: Detect breast cancer.
- Pap Smear: Detects cervical cancer.
- Stool Occult Blood Test: Detects colon cancer.
- Blood Pressure Checks: Detects hypertension.
- PSA Test: Detects prostate cancer.



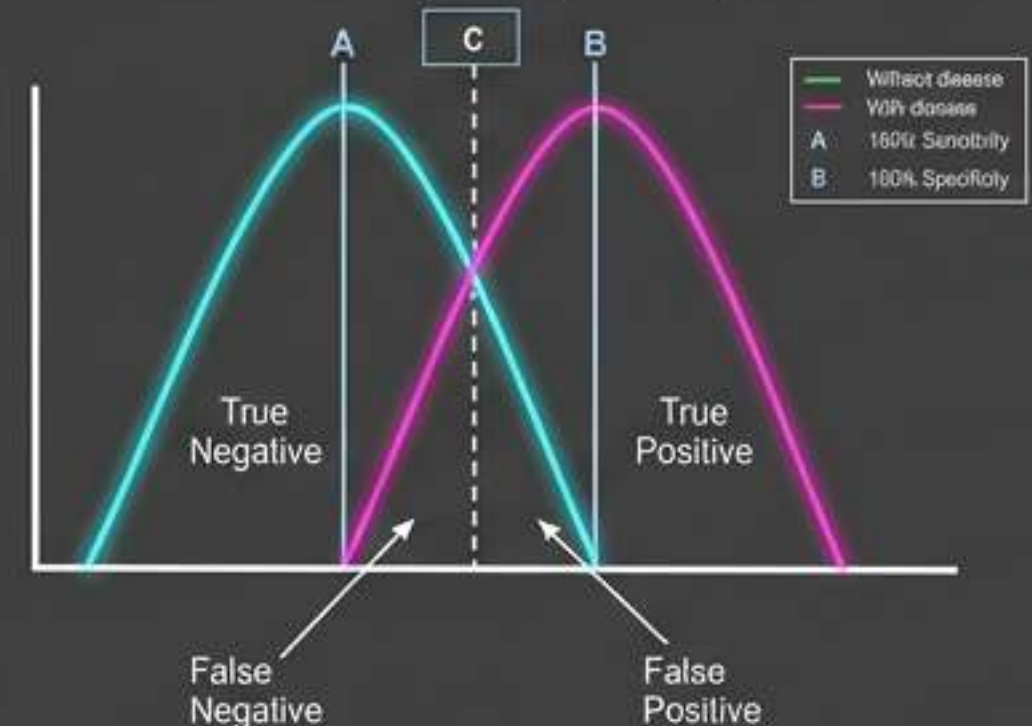
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CALIBRATION: THE TRADE-OFF

Sensitivity: Ability to correctly identify those with the disease (True Positives).

Specificity: Ability to correctly identify those without the disease (True Negatives).

Sensitivity vs. Specificity





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THE AMBUSH: EARLY TREATMENT

Definition: The immediate follow-up to screening.

Real World Example: Screening a pregnant woman for anemia and providing iron supplements immediately.

Outcome: Reduces suffering and prevents the disease from advancing to the clinical stage.



Sensei...
we missed some.
The Vector manifested
fully in Sector 4. The
damage is done.



Do not despair.
The disease has
occurred, but our duty
remains. Now we
initiate Tertiary
Prevention.



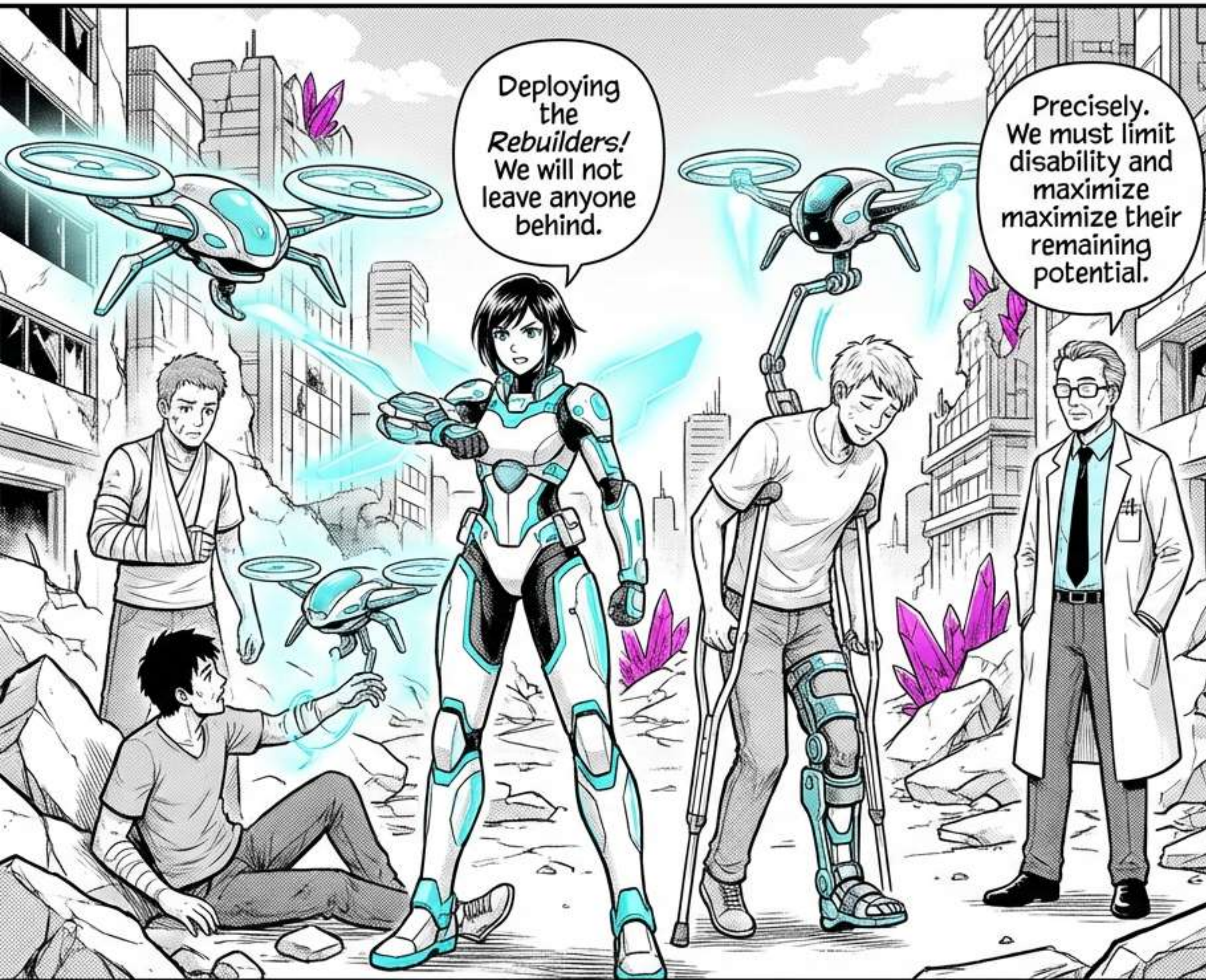
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THE DAMAGE: CLINICAL DISEASE

Stage: Clinical Disease.

Status: The disease has advanced past the early stages. The patient is symptomatic, and the illness is established.

Shift in Strategy:
From "prevention of occurrence"
to "prevention of impact."



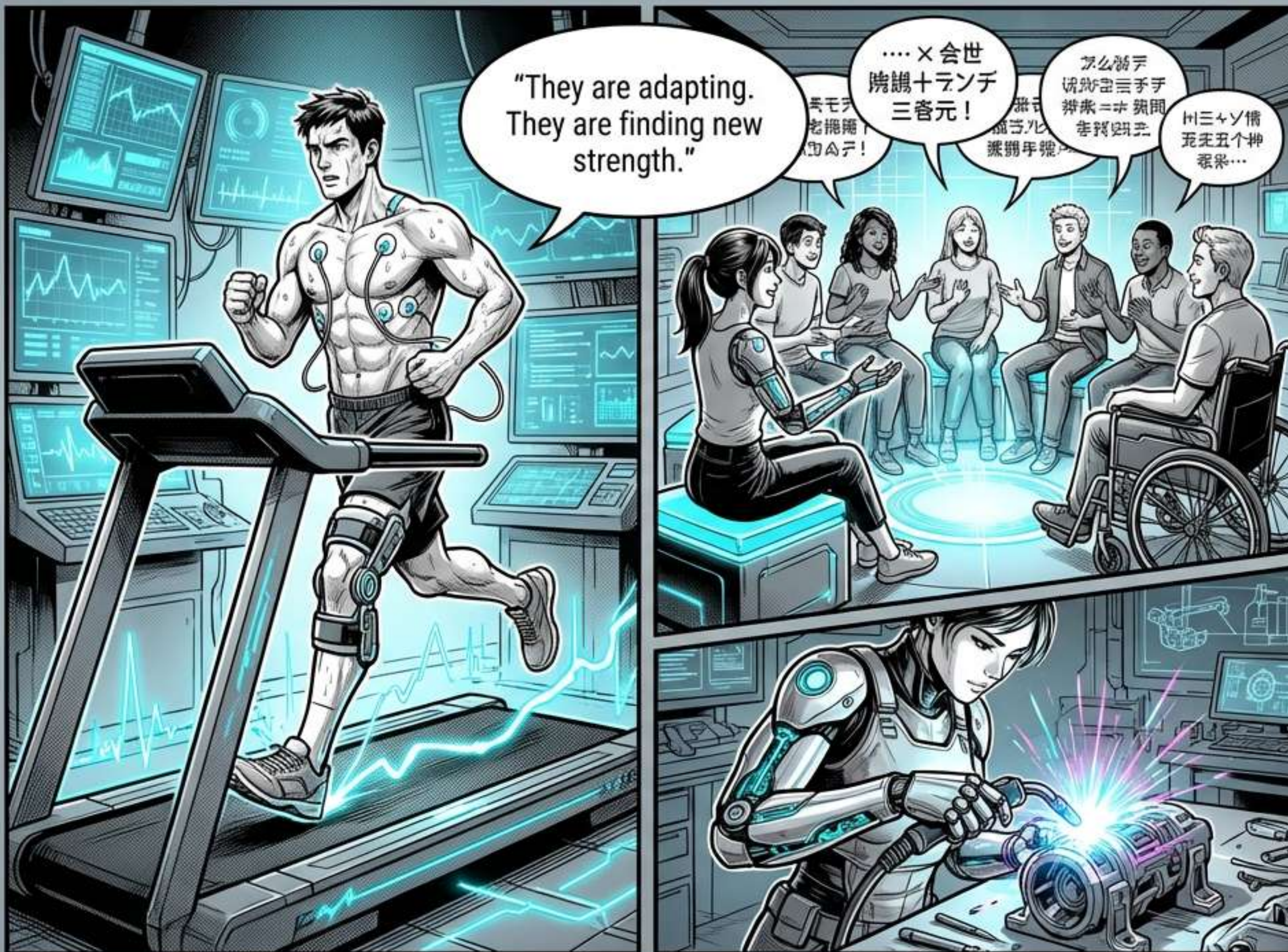
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SHIELD 3: TERTIARY PREVENTION

Definition: Reducing the impact of established disease.

Goals:

1. Eliminate or reduce disability.
2. Minimize suffering.
3. Maximize potential years of quality life.
4. Avoid dependency on others.



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REHABILITATION & RECOVERY

Components of Tertiary Prevention:

- **Rehabilitation:** Cardiac rehab after bypass surgery; Physical therapy.
- **Chronic Management:** Managing diabetes complications.
- **Support Groups:** Improving coping skills (e.g., for depression).
- **Vocational Training:** Job placement for those with acquired disabilities.



THE CYCLE OF DEFENSE: SUMMARY

PRIMARY PREVENTION

- **Goal:** Prevent Onset.
- **Action:** Immunization, Diet, Hygiene.

SECONDARY PREVENTION

- **Goal:** Early Detection.
- **Action:** Screening, Self-exams.

TERTIARY PREVENTION

- **Goal:** Improve Quality of Life.
 - **Action:** Rehabilitation, Management.
-
- **Key Takeaway:** Prevention is always better than a cure.



FIELD NOTES

To master the art of the Three Shields, consult the sacred text:

Epidemiology For Dummies by Amal K. Mitra, MD, DrPH.

Final Thought:

"Public health is a multidisciplinary field whose goal is to promote the health of a population through organized community efforts."